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Entity relation strength identification using spatiotemporal data

Abstract

A method includes obtaining information associated with multiple collocation events. Each collocation event is associated with an occurrence where multiple entities were collocated with one another. The information associated with each collocation event identifies the entities associated with the collocation event, a duration of the collocation event, one or more locations associated with the collocation event, and a time associated with the collocation event. The method also includes identifying strengths of different relationships involving different ones of the entities based on the information. The strengths are determined using one or more features of the collocation events.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11620338	12/2022	Bullard	707/798	G06F 16/287
2005/0075917	12/2004	Flores	705/7.11	G06Q 10/06
2010/0146118	12/2009	Wie	709/227	G06Q 10/10
2014/0019533	12/2013	Sherman	709/204	G06Q 10/10
2014/0074545	12/2013	Minder	705/7.27	G06Q 10/0633
2017/0111462	12/2016	Oberli et al.	N/A	N/A
2017/0236081	12/2016	Grady Smith	705/7.36	G06Q 10/0637
2017/0255868	12/2016	Gelinas et al.	N/A	N/A
2019/0130359	12/2018	Sterczyk	N/A	G06Q 10/105
2020/0104337	12/2019	Kelly et al.	N/A	N/A
2020/0364588	12/2019	Knox	N/A	G06V 40/20
2021/0065201	12/2020	Hudson	N/A	G06F 16/284

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2020167316	12/2019	WO	N/A

OTHER PUBLICATIONS

Hanson, "Using Scikit-learn's Binary Trees to Efficiently Find Latitude and Longitude Neighbors," Towards Data Science, Sep. 2020, 3 pages. cited by applicant

International Search Report and Written Opinion of the International Searching Authority dated Sep. 27, 2023 in connection with International Patent Application No. PCT/US23/63597, 8 pages. cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) This disclosure is generally directed to data processing systems. More specifically, this disclosure is directed to entity relation strength identification using spatiotemporal data.

BACKGROUND

(2) Various applications involve attempting to determine whether two or more entities (such as two or more people) have some type of relationship with one another. For example, business networking applications may attempt to determine whether various business professionals are familiar with one another and meet one another from time to time. As another example, intelligence applications may attempt to determine who various surveillance targets interact with, how often they interact, and when and where they interact. Often times, these types of applications rely on rules-based queries to extract information that is collected from specific sources of information.

SUMMARY

(3) This disclosure relates to entity relation strength identification using spatiotemporal data.

(4) In a first embodiment, a method includes obtaining information associated with multiple collocation events. Each collocation event is associated with an occurrence where multiple entities were collocated with one another. The information associated with each collocation event identifies the entities associated with the collocation event, a duration of the collocation event, one or more locations associated with the collocation event, and a time associated with the collocation event. The method also includes identifying strengths of different relationships involving different ones of the entities based on the information. The strengths are determined using one or more features of the collocation events.

(5) In a second embodiment, an apparatus includes at least one processing device configured to obtain information associated with multiple collocation events. Each collocation event is associated with an occurrence where multiple entities were collocated with one another. The information associated with each collocation event identifies the entities associated with the collocation event, a duration of the collocation event, one or more locations associated with the collocation event, and a time associated with the collocation event. The at least one processing device is also configured to identify strengths of different relationships involving different ones of the entities based on the information. The strengths are based on one or more features of the collocation events.

(6) In a third embodiment, a non-transitory computer readable medium stores computer readable program code that when executed causes one or more processors to obtain information associated with multiple collocation events. Each collocation event is associated with an occurrence where multiple entities were collocated with one another. The information associated with each collocation event identifies the entities associated with the collocation event, a duration of the collocation event, one or more locations associated with the collocation event, and a time associated with the collocation event. The medium also stores computer readable program code that when executed causes the one or more processors to identify strengths of different relationships involving different ones of the entities based on the information. The strengths are based on one or more features of the collocation events.

(7) Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of this disclosure, reference is now made to the following description, taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 illustrates an example system supporting entity relation strength identification using spatiotemporal data according to this disclosure;

(3) FIG. 2 illustrates an example device supporting entity relation strength identification using spatiotemporal data according to this disclosure;

(4) FIG. 3 illustrates an example architecture supporting entity relation strength identification using spatiotemporal data according to this disclosure;

(5) FIGS. 4 through 7 illustrate an example technique for entity relation strength identification using spatiotemporal data according to this disclosure; and

(6) FIG. 8 illustrates an example method for entity relation strength identification using spatiotemporal data according to this disclosure.

DETAILED DESCRIPTION

(7) FIGS. 1 through 8, described below, and the various embodiments used to describe the principles of the present disclosure are by way of illustration only and should not be construed in any way to limit the scope of this disclosure. Those skilled in the art will understand that the principles of the present disclosure may be implemented in any type of suitably arranged device or system.

(8) As noted above, various applications involve attempting to determine whether two or more entities (such as two or more people) have some type of relationship with one another. For example, business networking applications may attempt to determine whether various business professionals are familiar with one another and meet one another from time to time. As another example, intelligence applications may attempt to determine who various surveillance targets interact with, how often they interact, and when and where they interact. Often times, these types of applications rely on rules-based queries to extract information that is collected from specific sources of information.

(9) Unfortunately, conventional approaches for identifying relationships involving entities can suffer from various shortcomings. For example, rules-based queries are typically slow, require manual effort, and do not scale well as the amount of data to be processed increases. Also, it is generally difficult to automate and scale correlations of multiple data streams, and it is often difficult to integrate new sources of data into these approaches. Further, while these approaches may be able to identify whether certain entities have relationships, these approaches cannot quantify the strengths of those relationships. The strengths of relationships may be useful in various scenarios, such as when large datasets of many relations are being processed.

(10) This disclosure provides for entity relation strength identification using spatiotemporal data. As described in more detail below, collocation events can be obtained, such as based on information from one or more data sources. Each collocation event identifies an event or occurrence where multiple entities were collocated with one another. For example, each collocation event may identify an occurrence where two or more entities were at or near a common location, such as within a threshold distance of each other, for at least a threshold amount of time or for any amount of time. In some embodiments, each collocation event may identify the entities involved in the event, a duration of the event, one or more locations (such as a latitude and a longitude or other location information for the entities or for a related point) associated with the event, and a time (such as a start time and/or an end time and a date or other time-based information) of the event. The collocation events therefore represent spatiotemporal data, meaning data associated with both various locations and various times. A wide variety of data sources may be used here to obtain information used to generate or otherwise obtain collocation events, and data sources can be added and removed as needed.

(11) The collocation events are processed to identify relationships involving entities and strengths of those relationships. For example, to process the collocation events, various features may be identified using the collocation events, and these features may be subsequently processed to identify the relationships involving the entities and the strengths of those relationships. Any suitable features of the collocation events may be identified and used here. Examples of collocation event features that may be identified and processed could include the numbers of events (event frequencies) involving different entities, proximities or distances between or among entities during the events, and durations of the events. Depending on the implementation, these features may be determined for a single period of time or for multiple periods of time. Examples of timeframes that may be used here could include seven days, thirty days, and/or ninety days. The identified

collocation event features can also be scaled so that the feature values of those event features are expressed relative to a common scale or range, which can facilitate more effective aggregation of the identified collocation event features.

(12) The identified collocation event features may be processed in any suitable manner to identify the relationships involving the entities and the strengths of those relationships. For example, the features may be weighted differently and combined or otherwise processed, used as inputs to one or more machine learning models, analyzed using one or more statistical techniques, or used in any other suitable manner. The results of the feature analysis or analyses may include relationship scores, which can be used as indicators of both (i) identified relationships involving specified entities and (ii) the strengths of the identified relationships. Since the analysis can be repeated over time (such as daily), this can allow sequences of relationship scores to be determined for different entities over time.

(13) The ability to identify the strengths of relationships that are determined using spatiotemporal data can help to provide more effective differentiation of various types of relationships among entities and provide better resolution of those entities' relationships. The quantified relationships can also enable various downstream processing tasks, such as identifying key related entities given a specific entity of interest. In other words, a downstream processing task may be configured to identify, for a specific identified entity, which other entity or entities have key or strongest relationships with the specific identified entity. As a particular example, the quantified relationships can be used as input features into one or more machine learning models that are used in downstream processing tasks.

(14) FIG. 1 illustrates an example system **100** supporting entity relation strength identification using spatiotemporal data according to this disclosure. For example, the system **100** shown here may be used to identify relationships involving entities and the strengths of those relationships as described in more detail below. As shown in FIG. 1, the system **100** includes user devices **102a-102d**, one or more networks **104**, one or more application servers **106**, and one or more database servers **108** associated with one or more databases **110**. Each user device **102a-102d** communicates over the network **104**, such as via a wired or wireless connection. Each user device **102a-102d** represents any suitable device or system used by at least one user to provide or receive information, such as a desktop computer, a laptop computer, a smartphone, and a tablet computer. However, any other or additional types of user devices may be used in the system **100**.

(15) The network **104** facilitates communication between various components of the system **100**. For example, the network **104** may communicate Internet Protocol (IP) packets, frame relay frames, Asynchronous Transfer Mode (ATM) cells, or other suitable information between network addresses. The network **104** may include one or more local area networks (LANs), metropolitan area networks (MANs), wide area networks (WANs), all or a portion of a global network such as the Internet, or any other communication system or systems at one or more locations. In some cases, the network **104** may represent a private or internal network used by an organization that is processing data for entity relation strength identification.

(16) The application server **106** is coupled to the network **104** and is coupled to or otherwise communicates with the database server **108**. The application server **106** performs various operations to support entity relation strength identification using spatiotemporal data. For example, the application server **106** may execute one or more applications **112**, which can be used to implement the techniques for entity relation strength identification. The one or more applications **112** here can use data from and/or store data in the database **110** via the database server **108** during operation. Note that the database server **108** may also be used within the application server **106** to store information, in which case the application server **106** itself may store the information used to support entity relation strength identification. The results of the entity relation strength identification operations may be used in any suitable manner. For instance, identified relationships involving entities and strengths of the relationships (or information based thereon) may be

presented to one or more users, such as via one or more user devices **102a-102d**.

(17) The database server **108** operates to store and facilitate retrieval of various information used, generated, or collected by the application server **106** and the user devices **102a-102d** in the database **110**. For example, the database server **108** may store various information used to identify the relationships involving entities and the strengths of those relationships. At least some of the data used to perform the entity relation strength identification operations may be obtained via one or more external networks **114**, such as the Internet. For instance, one or more external data sources **116a-116n** may be coupled to or otherwise communicate over the one or more external networks **114** and provide information to the application server **106** for use and/or to the database server **108** for storage in the database **110**. The one or more external data sources **116a-116n** may include any suitable number(s) and type(s) of data source(s) used for entity relation strength identification, such as any suitable source of spatiotemporal data related to one or multiple entities.

(18) Although FIG. **1** illustrates one example of a system **100** supporting entity relation strength identification using spatiotemporal data, various changes may be made to FIG. **1**. For example, the system **100** may include any number of user devices **102a-102d**, networks **104**, **114**, application servers **106**, database servers **108**, databases **110**, and data sources **116a-116n**. Also, these components may be located in any suitable locations and might be distributed over a large area. In addition, while FIG. **1** illustrates one example operational environment in which entity relation strength identification may be used, this functionality may be used in any other suitable system.

(19) FIG. **2** illustrates an example device **200** supporting entity relation strength identification using spatiotemporal data according to this disclosure. One or more instances of the device **200** may, for example, be used to at least partially implement the functionality of the application server **106** of FIG. **1**. However, the functionality of the application server **106** may be implemented in any other suitable manner. In some embodiments, the device **200** shown in FIG. **2** may form at least part of a user device **102a-102d**, application server **106**, or database server **108** in FIG. **1**. However, each of these components may be implemented in any other suitable manner.

(20) As shown in FIG. **2**, the device **200** denotes a computing device or system that includes at least one processing device **202**, at least one storage device **204**, at least one communications unit **206**, and at least one input/output (I/O) unit **208**. The processing device **202** may execute instructions that can be loaded into a memory **210**. The processing device **202** includes any suitable number(s) and type(s) of processors or other processing devices in any suitable arrangement. Example types of processing devices **202** include one or more microprocessors, microcontrollers, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or discrete circuitry.

(21) The memory **210** and a persistent storage **212** are examples of storage devices **204**, which represent any structure(s) capable of storing and facilitating retrieval of information (such as data, program code, and/or other suitable information on a temporary or permanent basis). The memory **210** may represent a random access memory or any other suitable volatile or non-volatile storage device(s). The persistent storage **212** may contain one or more components or devices supporting longer-term storage of data, such as a read only memory, hard drive, Flash memory, or optical disc.

(22) The communications unit **206** supports communications with other systems or devices. For example, the communications unit **206** can include a network interface card or a wireless transceiver facilitating communications over a wired or wireless network, such as the network **104** or **114**. The communications unit **206** may support communications through any suitable physical or wireless communication link(s).

(23) The I/O unit **208** allows for input and output of data. For example, the I/O unit **208** may provide a connection for user input through a keyboard, mouse, keypad, touchscreen, or other suitable input device. The I/O unit **208** may also send output to a display, printer, or other suitable output device. Note, however, that the I/O unit **208** may be omitted if the device **200** does not require local I/O, such as when the device **200** represents a server or other device that can be

accessed remotely.

(24) Although FIG. 2 illustrates one example of a device **200** supporting entity relation strength identification using spatiotemporal data, various changes may be made to FIG. 2. For example, computing and communication devices and systems come in a wide variety of configurations, and FIG. 2 does not limit this disclosure to any particular computing or communication device or system.

(25) FIG. 3 illustrates an example architecture **300** supporting entity relation strength identification using spatiotemporal data according to this disclosure. For ease of explanation, the architecture **300** shown in FIG. 3 is described as being implemented on or supported by the application server **106** in the system **100** shown in FIG. 1, where the application server **106** may be implemented using one or more instances of the device **200** shown in FIG. 2. However, the architecture **300** shown in FIG. 3 could be implemented on or supported by any other suitable device(s) and in any other suitable system(s).

(26) As shown in FIG. 3, the architecture **300** generally operates to receive and process spatiotemporal data in the form of collocation events **302**. The collocation events **302** may be obtained in any suitable manner, such as from one or more data sources **116a-116n**. Each collocation event **302** identifies an occurrence in which multiple entities were collocated with one another, such as when two or more entities were within a threshold distance of each other for at least a threshold amount of time or for any amount of time. In some embodiments, each collocation event **302** may include information identifying the entities involved in the event, a duration of the event, one or more locations (such as a latitude and a longitude or other location information for the entities or for a related point) associated with the event, and a time (such as a start time and/or an end time and a date or other time-based information) of the event. Note that collocation events **302** associated with entities may represent data associated with the entities themselves and/or data associated with mobile devices (such as mobile smartphones or laptop/tablet computers) or other devices (such as vehicles) associated with the entities. Also note that the location(s) may represent the locations of the entities themselves or of their devices or a related location, such as a point between or close to the entities or a point around which the entities are located.

(27) In some embodiments, the collocation events **302** may be obtained via an API or other interface **304** and processed into a unified data format. The unified data format may represent a predefined format in which the collocation events **302** can be placed. As a particular example, the unified data format may use a specified data structure that includes fields for information such as entity identification, location(s) (like a latitude and a longitude or other location information) of the event, and a time (like a start time and/or an end time and a date or other time-based information) of the event, and duration. Any suitable interface **304** may be used here to obtain the collocation events **302**.

(28) Data processing operations **306** are performed to analyze the collocation events **302** in order to perform entity relation strength identification operations. Examples of how the data processing operations **306** may be performed to identify relationships and relationship strengths are provided below. The results of the data processing operations **306** may be used in any suitable manner. In this specific example, the processing results (or information based on the processing results) can be provided to one or more analysts **308**. As a particular example, the results of the data processing operations **306** may be provided via at least one graphical user interface presented by one or more user devices **102a-102d** used by the analyst(s) **308**.

(29) Note that the functions shown in or described with respect to FIG. 3 can be implemented in an electronic device, such as a computing device, in any suitable manner. For example, in some embodiments, at least some of the functions shown in or described with respect to FIG. 3 can be implemented or supported using one or more software applications or other software instructions that are executed by one or more processing devices of an application server **106**, device **200**, or other device. In other embodiments, at least some of the functions shown in or described with

respect to FIG. 3 can be implemented or supported using dedicated hardware components. In general, the functions shown in or described with respect to FIG. 3 can be performed using any suitable hardware or any suitable combination of hardware and software/firmware instructions.

(30) Although FIG. 3 illustrates one example of an architecture 300 supporting entity relation strength identification using spatiotemporal data, various changes may be made to FIG. 3. For example, identified relationships or relationship strengths may be used in any suitable manner.

(31) FIGS. 4 through 7 illustrate an example technique for entity relation strength identification using spatiotemporal data according to this disclosure. For ease of explanation, the technique shown in FIGS. 4 through 7 is described as being performed by the application server 106 in the system 100 shown in FIG. 1, where the application server 106 may be implemented using one or more instances of the device 200 shown in FIG. 2. As a particular example, the technique shown in FIGS. 4 through 7 may be implemented as part of the data processing operations 306 within the architecture 300 shown in FIG. 3. However, the technique shown in FIGS. 4 through 7 could be performed by any other suitable device(s) and in any other suitable system(s).

(32) As shown in FIG. 4, a feature identification process 400 obtains input data 402 in the form of collocation events. The collocation events in the input data 402 can include events that involve any suitable number of entities and that occur over any suitable range of locations. The collocation events in the input data 402 can also include events of various durations, and the collocation events in the input data 402 may span any suitable period of time. As a particular example, the collocation events in the input data 402 may represent the collocation events 302 described above.

(33) The events in the input data 402 may be divided into or otherwise associated with one or more timeframes 404. In this example, there are three timeframes 404 used, namely seven days, thirty days, and ninety days. Thus, events in the input data 402 that occurred within the last seven days may be placed within or otherwise associated with a first timeframe 404, events in the input data 402 that occurred within the last thirty days may be placed within or otherwise associated with a second timeframe 404, and events in the input data 402 that occurred within the last ninety days may be placed within or otherwise associated with a third timeframe 404. Thus, the same event may be included in multiple timeframes 404, and the number of events within each timeframe 404 will generally increase as the length of the timeframes 404 increases. Note, however, that the number and durations of the timeframes 404 shown here are for illustration only. In general, any suitable number of timeframes 404 may be used, and each timeframe 404 may be of any suitable duration. Also note that processing of collocation events can be repeatedly performed over time, so newer collocation events can be added to the timeframe(s) 404 and older collocation events can be dropped from the timeframe(s) 404 as time progresses.

(34) For each timeframe 404, a collection of features 406 for the collocation events within or otherwise associated with that timeframe 404 may be generated. Any suitable features of the collocation events may be determined for each timeframe 404 here. In this particular example, the features 406 include count features, entity distance features, and duration features. In some embodiments, these features 406 may be determined for pairs of entities identified in the input data 402 as having been collocated with one another. In other cases, these features 406 may be determined for two or more entities identified in the input data 402 as having been collocated with one another. This process effectively “featurizes” the collocation events in the input data 402 by producing features 406 defining the collocation events.

(35) In this example, each count feature 406 identifies (for two or more entities) the number of collocation events that involved those entities within a specified timeframe 404 and that had a specified duration or a duration that fell within a specific range. In this particular example, there are four count features 406, namely the number of collocation events in the timeframe 404 having a duration under a threshold (ten minutes), the number of collocation events in the timeframe 404 having a duration between two thresholds (ten minutes to thirty minutes), the number of collocation events in the timeframe 404 having a duration between two other thresholds (thirty minutes to sixty

minutes), and the number of collocation events in the timeframe **404** having a duration above a threshold (sixty minutes). Note, however, that the number and type of count features and the thresholds associated with the count features **406** can vary as needed or desired.

(36) Each entity distance feature **406** identifies (for two or more entities) a type of distance between or among those entities during collocation events involving those entities in the timeframe **404**. In this particular example, there are two entity distance features **406**, namely the minimum distance between or among multiple entities during collocation events involving those entities in the timeframe **404** and the average distance between or among those entities during collocation events involving those entities in the timeframe **404**. Note, however, that the number and type of entity distance features **406** can vary as needed or desired.

(37) Each duration feature **406** identifies (for two or more entities) a type of duration measurement associated with collocation events involving those entities in the timeframe **404**. In this particular example, there are three duration features **406**, namely the total duration of collocation events involving multiple entities in the timeframe **404**, the maximum duration of a collocation event involving those entities in the timeframe **404**, and the average duration of the collocation events involving those entities in the timeframe **404**. Note, however, that the number and type of duration features **406** can vary as needed or desired.

(38) Again, it should be noted that these features **406** can be determined for different combinations of entities (such as different pairs or groups of people) and for each timeframe **404** being used, and the features **406** can be calculated repeatedly over time (such as daily). As a result, these features **406** can be used to represent possible interactions of a large number of entities with one another over time. Also note that these features **406** may be determined in any suitable manner. In addition, it should be noted that the express identification of values for these features **406** may or may not occur in any given implementation. For example, in some embodiments, these features **406** may be expressly calculated and then processed as described below to identify entity relationships and relationship strengths. In other embodiments, the input data **402** itself (with or without division into timeframes **404**) may be processed directly, such as by one or more machine learning models, and used to identify entity relationships and relationship strengths.

(39) One common issue with calculated features **406** is that the features **406** often use different scales (such as frequencies or count numbers, distances, and times) and may use different numerical ranges to identify feature values. As a result, it can be difficult to aggregate the features **406** into a collection of values that can be processed together and used for identifying entity relationships and relationship strengths. For example, it may be difficult to weigh and combine features **406** or otherwise use features **406** having values that use different scales and that are expressed across different numerical ranges. Thus, feature values may optionally be pre-processed prior to further use in identifying entity relationships and relationship strengths.

(40) One example feature pre-processing technique is shown in FIG. 5, which illustrates that a scaling process **500** can take a set of feature values **502** as determined for one of the features **406** using one of the timeframes **404** and scale the feature values **502** to produce scaled feature values **504**. The feature values **502** here are represented as a distribution in graph form, where the horizontal axis represents feature values and the vertical axis represents a count of the number of feature values **502** within the timeframe **404** having each value. The scaled feature values **504** here are also represented as a distribution in graph form, where the horizontal axis represents scaled feature values and the vertical axis represents a count of the number of feature values **502** within the timeframe **404** having each scaled value. The feature values **502** here are scaled to have values between zero and one (inclusive).

(41) The function that is used to scale the feature values **502** of each feature **406** may be defined in any suitable manner to provide the desired scaling of those feature values **502** to a specified range. For example, a scaling function may be linear, non-linear, parametric, or non-parametric. Also, different scaling functions may be used for different features **406** when those features **406** use

higher and lower values to indicate different levels of association. For example, some features **406** (such as count features and duration features) may indicate stronger associations involving entities when values of those features are higher and weaker associations involving entities when values of those features are lower. In contrast, other features **406** (such as entity distance features) may indicate stronger associations involving entities when values of those features are lower and weaker associations involving entities when values of those features are higher. Thus, the scaling that is applied for values **502** of each feature **406** can be designed to support an appropriate modification of those values **502**. In some cases, the values **502** of all features **406** can be scaled so that higher values indicate stronger associations and lower values indicate weaker associations (or vice versa). (42) As a particular example of how scaling could be performed, it may be desirable to achieve improved resolution for a subset of feature values **502** related to one or more features **406**. For example, it may be desirable to obtain improved resolution of the top 20% or other subset of feature values **502** for the count and duration features **406** and to obtain improved resolution of the bottom 20% or other subset of feature values **502** for the entity distance features **406**. In these situations, the feature values **502** for the count and duration features **406** may be scaled so that the top 20% spans a wider range of values (since larger feature values **502** for these features **406** are most useful in identifying relationships and relationship strengths). Also, the feature values **502** for the entity distance features **406** may be scaled so that the bottom 20% spans a wider range of values (since smaller feature values **502** for those features **406** are most useful in identifying relationships and relationship strengths). In some embodiments, the scaling function applied to the feature values **502** of the count and duration features **406** or other features positively correlated with strength to produce the scaled feature values **504** may be defined as:

Feature Score = $1 - 0.2 \cdot \text{sup.}(\text{Feature Value} / 80\text{th})$

Here, Feature Value represents a feature value **502** being scaled, and Feature Score represents the scaled feature value **504** produced for that feature value **502**. Also, 80.sup.th represents a feature value **502** of the associated feature **406** at the 80.sup.th percentile of that feature **406**. Thus, this approach ensures that the bottom 80% of the feature values **502** for the count and duration features **406** occupy 20% of the scaled feature values **504**, while the top 20% of the feature values **502** for the count and duration features **406** are spread over 80% of the scaled feature values **504**. The scaling function used with the entity distance features **406** or other features negatively correlated with strength may be defined as the inverse of the scaling function defined above, which can be done to reorient the entity distance features **406** so that larger scaled feature values **504** indicate stronger associations. Thus, in some embodiments, the scaling function applied to the feature values **502** of the entity distance features **406** to produce the scaled feature values **504** may be defined as:

Feature Score = $0.2 \cdot \text{sup.}(\text{Feature Value} / 80\text{th})$

This allows the scaled feature values **504** for the entity distance features **406** to increase as entity distance decreases. Note, however, that these specific scaling functions are for illustration only and that other scaling functions may be used as needed or desired.

(43) As described above, the feature values **502** and **504** shown in FIG. 5 may represent the values determined for one of the features **406** in FIG. 4 using the input data **402** associated with one timeframe **404**. The process **500** shown in FIG. 5 may be repeated for each feature **406** determined for each timeframe **404** in order to scale all of the values **502** for each feature **406**. Doing this for all features **406** can help to align the ranges of values for the features **406** and enable those features **406** to be combined or otherwise processed further to identify entity relationships and relationship strengths.

(44) Again, it should be noted that the express scaling of the values **502** of the features **406** may or may not occur in any given implementation. For example, in some embodiments, the scaled feature values **504** may be expressly determined and then processed as described below to identify entity relationships and relationship strengths. In other embodiments, the input data **402** itself or the

features **406** and their feature values **502** themselves may be processed directly, such as by one or more machine learning models, and used to identify entity relationships and relationship strengths. (45) Once pre-processed or otherwise suitably obtained, the features **406** can be used to identify relationships and relationship strengths involving various entities. As noted above, there are various ways in which values of the features **406** can be used to identify relationships and relationship strengths. For example, the scaled feature values **504** may be weighted differently and combined or otherwise processed, used as inputs to one or more machine learning models, analyzed using one or more statistical techniques, or used in any other suitable manner. Several example approaches for using the scaled feature values **504** to identify relationships and relationship strengths are shown in FIG. **6**, which illustrates an example feature analysis process **600**.

(46) In some embodiments, the feature analysis process **600** relies on determining weighted feature values in order to identify the strengths of relationships involving entities. For example, as shown in FIG. **6**, different weights **602** can be defined for each of the timeframes **404**, and/or different weights **604** can be defined for each of the different types of features **406**. Note that these weights **602** and **604** are for illustration only and can vary as needed or desired. The weights **602** and **604** shown in FIG. **6** may be determined in any suitable manner. For instance, the weights **602** and **604** can be set manually or based on user configuration.

(47) In this example, the scaled feature values **504** of the features **406** within each timeframe **404** can be weighted using the appropriate weights **602**, and the scaled feature values **504** of the different types of features **406** within each timeframe **404** can also be weighted using the appropriate weights **604**. After weighting, the weighted feature values can be summed, processed using a maximum, minimum, averaging, median, percentile, or other function, or otherwise used to produce final scores **606**, which in this example are plotted in a graph. In this particular example, the final scores **606** for a pair or group of entities are plotted over time in the graph, where the total height of the plotted data for each day defines the total value of the final score **606** for that day. Larger final scores **606** can indicate stronger relationships, while smaller final scores **606** can indicate weaker relationships. The graph also illustrates how the different types of features **406** contribute to each final score **606**. Of course, the final scores **606** may be determined or presented in any other suitable manner. Note that the final scores **606** in the graph here relate to a specific pair or group of entities, and final scores **606** may be similarly determined for each of multiple pairs or groups of entities.

(48) In other embodiments, the final scores **606** may be determined directly without reference to explicit weights **602** and **604**. For instance, one or more machine learning models **608** may be used to process the input data **402**, the feature values **502** of the features **406**, the scaled feature values **504** of the features **406**, and/or other data in order to generate the final scores **606**. In those embodiments, the one or more machine learning models **608** may be implemented in any suitable manner. For instance, one or more machine learning models **608** may be generated using semi-supervised learning and/or supervised learning with labeled training data, where the labeled training data is used to train the one or more machine learning models **608** to generate predictions (final scores **606**) based on data provided to the model(s) **608**. One or more machine learning models **608** may also be derived from an unsupervised approach, such as by using anomaly detection. The one or more machine learning models **608** may be used here to generate suitable final scores **606**, and the final scores **606** may or may not identify the contributions of different types of features **406** to the final scores **606**.

(49) In still other embodiments, computations other than weighted processing can be used to generate the final scores **606**. These computations may be used to process the input data **402**, the feature values **502** of the features **406**, the scaled feature values **504** of the features **406**, and/or other data in order to generate the final scores **606**. As a particular example, feature values **502** or **504** or other data may be processed using one or more statistical techniques, such as percentiles, z-scores, p-values, or mean absolute deviations of distributions, to identify the final scores **606**.

Again, the final scores **606** may or may not identify the contributions of different types of features **406** to the final scores **606**.

(50) In whatever manner the final scores **606** are determined, the final scores **606** here can be used as an indicator of (i) whether a relationship exists involving two or more entities and (ii) the strength of that relationship, if any. For example, smaller final scores **606** can be indicative of a lack of relationship involving entities or a weak relationship involving entities, while larger final scores **606** can be indicative of a stronger relationship involving entities. Moreover, because the final scores **606** can be determined over time, the final scores **606** can be used to measure trends over time, such as whether a relationship strengthens or weakens over time, is periodic, etc. In this way, the architecture **300** is able to provide a quantitative measure of the strengths of relationships involving various entities, which can be useful during various downstream processing tasks.

(51) A summary of the process for generating the final scores **606** using weighted processing is shown in FIG. 7. As shown in FIG. 7, the feature values **502** can be generated for a number of features **406** within each of multiple timeframes **404**. Scaling can be performed to convert the feature values **502** of the features **406** into scaled feature values **504**. For instance, the scaled feature values **504** may be defined as percentages or other values within a scale ranging from zero to one. The weights **602** can be applied to the scaled feature values **504** in order to generate intermediate values **702**, where each intermediate value **702** is determined by applying different weights **602** to the scaled feature values **504** in different timeframes **404**. The weights **604** can be applied to the intermediate values **702** in order to generate the final score **606** for a specific pair or group of entities. This can be repeated for any number of pairs or groups of entities in order to identify the strengths of relationships involving those entities. Note that it is also possible to compute the final score **606** directly from the scaled feature values **504** using the weights **602** and **604** (without separately calculating the intermediate values **702**). Also note that the final score **606** can be determined repeatedly over time in order to generate multiple final scores **606** identifying the strength of the relationship over time.

(52) Although FIGS. 4 through 7 illustrate one example of a technique for entity relation strength identification using spatiotemporal data, various changes may be made to FIGS. 4 through 7. For example, the actual values used in FIGS. 4 through 7 (such as timeframes, durations, feature values, feature value distributions, weights, and final scores) can easily vary based on the implementation and the specific input data being processed. Also, the graphs shown here are for illustration and explanation only.

(53) Note that while the embodiments using weighted processing of feature values have been described above as using common weights for all values of the same feature, it is possible to weight values of the same feature differently depending on the circumstances. For example, with respect to the event frequency features, it may be true that more frequent occurrences involving entities may be indicative of a relationship involving those entities. However, there may be some situations where that is not necessarily the case. As a particular example, two people who routinely visit the same coffee shop multiple times a week at around the same time each workday may simply represent two people who get coffee at the same location before going to work. However, two people who fly in from out of town and then meet at the same coffee shop multiple times within a timeframe **404** may be much more indicative of a relationship involving those people. It is therefore possible to have different weights for the same types of feature values, where those weights may be dependent on other factors. The same or similar types of weighting adjustments may be used for any of the features of the collocation events described above, where the weights can vary based on other factors not necessarily captured by those features themselves. In addition to or in lieu of weighting different values of the same feature differently depending on the circumstances, separate features may be used to differentiate the collocation events described above, such as when one or more other features consider the distance(s) traveled by one or more entities to a collocation event and/or the distance(s) traveled by one or more entities following a collocation

event.

(54) FIG. 8 illustrates an example method **800** for entity relation strength identification using spatiotemporal data according to this disclosure. For ease of explanation, the method **800** shown in FIG. 8 is described as being performed using the application server **106** in the system **100** shown in FIG. 1, where the application server **106** may be implemented using one or more instances of the device **200** shown in FIG. 2. As a particular example, the method **800** shown in FIG. 8 may be performed as part of the data processing operations **306** within the architecture **300** shown in FIG. 3. However, the method **800** shown in FIG. 8 could be performed using any other suitable device(s) and in any other suitable system(s).

(55) As shown in FIG. 8, information identifying collocation events is obtained at step **802**. This may include, for example, the processing device **202** of the application server **106** obtaining information from one or more data sources **116a-116n**, where the information may represent collocation events involving any number of entities. One or more timeframes to be used when analyzing the collocation events are identified at step **804**. This may include, for example, the processing device **202** of the application server **106** identifying one or more timeframes **404** to be used when processing the collocation events.

(56) Features of the collocation events occurring within each timeframe are identified at step **806**. This may include, for example, the processing device **202** of the application server **106** processing the input data **402** to identify values **502** of features **406** associated with the collocation events. As a particular example, this may include the processing device **202** of the application server **106** processing the input data **402** to identify count, entity distance, and duration features for different pairs or groups of entities associated with the collocation events. The feature values of the features can be scaled at step **808**. This may include, for example, the processing device **202** of the application server **106** processing the feature values **502** of various features **406** in order to (among other things) scale the feature values **502** to a common scale and produce scaled feature values **504** for those features **406**.

(57) The feature values are used to identify scores related to the strengths of relationships involving entities at step **810**. This may include, for example, the processing device **202** of the application server **106** weighting the feature values **502** or **504** differently and combining or otherwise processing the weighted feature values, providing the feature values **502** or **504** as inputs to one or more machine learning models **608**, analyzing the feature values **502** or **504** using one or more statistical techniques, or processing the feature values **502** or **504** in any other suitable manner to produce the final scores **606**. The final scores **606** can be used as indicators of whether relationships exist involving certain entities and the strengths of those relationships.

(58) The scores may be stored, output, or used in some manner at step **812**. This may include, for example, the processing device **202** of the application server **106** including one or more of the final scores **606** in a graphical user interface or report. This may also or alternatively include providing at least one of the final scores **606** to a downstream processing task, such as a task that identifies the key or strongest entity relationship(s) associated with a specific entity. This may also or alternatively include storing the final scores **606** in the database **110** for further processing. In general, the final scores **606** or other indicators of relationship strengths may be used in any suitable manner.

(59) Although FIG. 8 illustrates one example of a method **800** for entity relation strength identification using spatiotemporal data, various changes may be made to FIG. 8. For example, while shown as a series of steps, various steps in FIG. 8 may overlap, occur in parallel, occur in a different order, or occur any number of times. As a particular example, various steps may be repeated over time to calculate multiple scores for the same entities over time. Also, as noted above, some embodiments of this disclosure may process input data **402** directly, process feature values **502** without scaling, or otherwise process data to identify relationship strengths, in which case various steps in FIG. 8 may be omitted or altered.

(60) Note that while it may be assumed in some instances above that the described approaches are being used to identify relationships and relationship strengths involving different people, the same or similar types of processing may be performed to identify relationships and relationship strengths involving different groups or other collections of people. For example, it is possible to identify relationships and relationship strengths involving different individuals, identify individuals associated with different groups, and combine the final scores associated with any relationships involving people of different groups. This could be done to provide indications of the strengths of relationships involving the different groups.

(61) The following describes example embodiments of this disclosure that implement entity relation strength identification using spatiotemporal data. However, other embodiments may be used in accordance with the teachings of this disclosure.

(62) In a first embodiment, a method includes obtaining information associated with multiple collocation events. Each collocation event is associated with an occurrence where multiple entities were collocated with one another. The information associated with each collocation event identifies the entities associated with the collocation event, a duration of the collocation event, one or more locations associated with the collocation event, and a time associated with the collocation event. The method also includes identifying strengths of different relationships involving different ones of the entities based on the information. The strengths are determined using one or more features of the collocation events.

(63) In a second embodiment, an apparatus includes at least one processing device configured to obtain information associated with multiple collocation events. Each collocation event is associated with an occurrence where multiple entities were collocated with one another. The information associated with each collocation event identifies the entities associated with the collocation event, a duration of the collocation event, one or more locations associated with the collocation event, and a time associated with the collocation event. The at least one processing device is also configured to identify strengths of different relationships involving different ones of the entities based on the information. The strengths are based on one or more features of the collocation events.

(64) In a third embodiment, a non-transitory computer readable medium stores computer readable program code that when executed causes one or more processors to obtain information associated with multiple collocation events. Each collocation event is associated with an occurrence where multiple entities were collocated with one another. The information associated with each collocation event identifies the entities associated with the collocation event, a duration of the collocation event, one or more locations associated with the collocation event, and a time associated with the collocation event. The medium also stores computer readable program code that when executed causes the one or more processors to identify strengths of different relationships involving different ones of the entities based on the information. The strengths are based on one or more features of the collocation events.

(65) Any single one or any suitable combination of the following features may be used with the first, second, or third embodiment. The one or more features of the collocation events may include one or more of (i) one or more count features associated with one or more frequencies or numbers of collocation events involving specified ones of the entities, (ii) one or more distance features associated with one or more distances between or among the specified entities during the collocation events, and (iii) one or more duration features associated with one or more lengths of collocation involving the specified entities during the collocation events. The one or more features of the collocation events may be identified for each of one or more timeframes, and the strengths of the different relationships may be determined using or may be based on the one or more features of the collocation events during the one or more timeframes. Feature values of at least one of the one or more features may be scaled, and the strengths of the different relationships may be determined using or may be based on the scaled feature values of the at least one of the one or more features. The feature values of the at least one of the one or more features may be scaled such that the

feature values of all features span a common range of values. A first one of the one or more features may use lower values to identify weaker associations involving entities and higher values to identify stronger associations involving entities, a second one of the one or more features may use lower values to identify stronger associations involving entities and higher values to identify weaker associations involving entities, and one of the first and second features may be scaled to reverse the values representing stronger and weaker associations involving entities. The strengths of the different relationships may be identified by (i) for each relationship, determining an aggregation of feature values associated with the entities involved in that relationship, (ii) using at least one machine learning model trained to identify the strengths of the relationships, and/or (iii) using one or more statistical calculations to identify the strengths of the relationships. The strengths of the different relationships may be identified by, for each relationship, determining an aggregation of feature values associated with the entities involved in that relationship, where unequal weights are associated with the feature values of each of at least one of the one or more features of the collocation events. At least one of the one or more features may be associated with distances traveled by entities to or from collocation events.

(66) In some embodiments, various functions described in this patent document are implemented or supported by a computer program that is formed from computer readable program code and that is embodied in a computer readable medium. The phrase “computer readable program code” includes any type of computer code, including source code, object code, and executable code. The phrase “computer readable medium” includes any type of medium capable of being accessed by a computer, such as read only memory (ROM), random access memory (RAM), a hard disk drive (HDD), a compact disc (CD), a digital video disc (DVD), or any other type of memory. A “non-transitory” computer readable medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other signals. A non-transitory computer readable medium includes media where data can be permanently stored and media where data can be stored and later overwritten, such as a rewritable optical disc or an erasable storage device.

(67) It may be advantageous to set forth definitions of certain words and phrases used throughout this patent document. The terms “application” and “program” refer to one or more computer programs, software components, sets of instructions, procedures, functions, objects, classes, instances, related data, or a portion thereof adapted for implementation in a suitable computer code (including source code, object code, or executable code). The term “communicate,” as well as derivatives thereof, encompasses both direct and indirect communication. The terms “include” and “comprise,” as well as derivatives thereof, mean inclusion without limitation. The term “or” is inclusive, meaning and/or. The phrase “associated with,” as well as derivatives thereof, may mean to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, have a relationship to or with, or the like. The phrases “at least one of” and “one or more of,” when used with a list of items, means that different combinations of one or more of the listed items may be used, and only one item in the list may be needed. For example, “at least one of: A, B, and C” includes any of the following combinations: A, B, C, A and B, A and C, B and C, and A and B and C.

(68) The description in the present disclosure should not be read as implying that any particular element, step, or function is an essential or critical element that must be included in the claim scope. The scope of patented subject matter is defined only by the allowed claims. Moreover, none of the claims invokes 35 U.S.C. § 112(f) with respect to any of the appended claims or claim elements unless the exact words “means for” or “step for” are explicitly used in the particular claim, followed by a participle phrase identifying a function. Use of terms such as (but not limited to) “mechanism,” “module,” “device,” “unit,” “component,” “element,” “member,” “apparatus,” “machine,” “system,” “processor,” or “controller” within a claim is understood and intended to refer to structures known to those skilled in the relevant art, as further modified or enhanced by the

features of the claims themselves, and is not intended to invoke 35 U.S.C. § 112(f).

(69) While this disclosure has described certain embodiments and generally associated methods, alterations and permutations of these embodiments and methods will be apparent to those skilled in the art. Accordingly, the above description of example embodiments does not define or constrain this disclosure. Other changes, substitutions, and alterations are also possible without departing from the spirit and scope of this disclosure, as defined by the following claims.

Claims

1. A method comprising: obtaining information associated with multiple collocation events from multiple data sources, wherein each collocation event is associated with an occurrence where multiple entities were collocated with one another; pre-processing the information from the multiple data sources into a unified data format using a first machine learning model, the unified data format defining: the multiple entities associated with a collocation event; a duration of the collocation event; one or more locations associated with the collocation event; and a time associated with the collocation event; determining, by a second machine learning model, multiple features of the multiple collocation events using the information in the unified data format, wherein the second machine learning model is distinct from the first machine learning model; providing the multiple determined features of the multiple collocation events to a third machine learning model, wherein the third machine learning model is distinct from the first machine learning model and the second machine learning model; and identifying strengths of different relationships involving different ones of the multiple entities using the third machine learning model based on the multiple determined features of the multiple collocation events; wherein the third machine learning model is trained to identify, using the information in the unified data format, the strengths of the different relationships by generating scores based on the multiple determined features of the multiple collocation events.
2. The method of claim 1, wherein the multiple determined features of the multiple collocation events comprise: one or more count features associated with one or more frequencies or numbers of collocation events involving specified entities of the multiple entities; one or more distance features associated with one or more distances between or among the specified entities during the multiple collocation events; and one or more duration features associated with one or more lengths of collocation involving the specified entities during the multiple collocation events.
3. The method of claim 2, wherein: the multiple determined features of the multiple collocation events are determined for each of multiple timeframes; and the strengths of the different relationships are determined using the multiple determined features of the multiple collocation events during the multiple timeframes.
4. The method of claim 1, further comprising: scaling feature values of at least one of the multiple determined features to generate scaled features of the at least one of the multiple determined features; wherein the strengths of the different relationships are determined using the scaled feature values of the at least one of the multiple determined features.
5. The method of claim 4, wherein the scaled feature values of all of the determined features span a common range of values.
6. The method of claim 4, wherein: a first determined feature of the multiple determined features uses values below a first threshold to identify weaker associations involving the multiple entities and values above the first threshold to identify stronger associations involving the multiple entities, wherein a strength of an association reflects a quantification of a grouping of entities of the multiple entities; a second determined feature of the multiple determined features uses values below a second threshold to identify the stronger associations involving the multiple entities and values above the second threshold to identify the weaker associations involving the multiple entities; and scaling the feature values comprises scaling the feature values of the first determined

feature or the second determined feature to reverse values representing the stronger associations involving the multiple entities and the weaker associations involving the multiple entities.

7. The method of claim 1, wherein identifying the strengths of the different relationships comprises, for each of the different relationships, determining an aggregation of feature values associated with entities among multiple entities involved in a relationship.

8. The method of claim 1, wherein the second machine learning model is trained during unsupervised learning to perform anomaly detection.

9. The method of claim 1, wherein identifying the strengths of the different relationships comprises using one or more statistical calculations to identify the strengths of the different relationships.

10. The method of claim 1, wherein the second machine learning model is trained during semi-supervised or supervised learning using labeled training data.

11. The method of claim 1, wherein: identifying the strengths of the different relationships comprises, for each of the different relationships, determining an aggregation of feature values associated with multiple entities involved in a relationship; and unequal weights are associated with feature values of each determined feature of the multiple collocation events.

12. The method of claim 11, wherein: the multiple collocation events comprise meetings between a plurality of entities at a specific location; the feature values include feature values associated with a frequency at which the meetings at the specific location occur; the feature values associated with the frequency at which the meetings at the specific location occur have unequal weights; and the unequal weights are based on different distances that different entities of the plurality of entities traveled to the specific location.

13. The method of claim 1, wherein at least one of the multiple determined features is associated with distances traveled by one or more of the multiple entities to or from one or more of the multiple collocation events.

14. An apparatus comprising: at least one processing device configured to: obtain information associated with multiple collocation events from multiple data sources, wherein each collocation event is associated with an occurrence where multiple entities were collocated with one another; pre-process the information from the multiple data sources into a unified data format using a first machine learning model, the unified data format defining: the multiple entities associated with a collocation event; a duration of the collocation event; one or more locations associated with the collocation event; and a time associated with the collocation event; determine, by a second machine learning model, multiple features of the multiple collocation events using the information in the unified data format, wherein the second machine learning model is distinct from the first machine learning model; provide the multiple determined features of the multiple collocation events to a third machine learning model, wherein the third machine learning model is distinct from the first machine learning model and the second machine learning model; and identify strengths of different relationships involving different ones of the multiple entities using the third machine learning model based on the multiple determined features of the multiple collocation events; wherein the third machine learning model is trained to identify, using the information in the unified data format, the strengths of the different relationships by generating scores based on the multiple determined features of the multiple collocation events.

15. The apparatus of claim 14, wherein the multiple determined features of the multiple collocation events comprise: one or more count features associated with one or more frequencies or numbers of collocation events involving specified entities of the multiple entities; one or more distance features associated with one or more distances between or among the specified entities during the multiple collocation events; and one or more duration features associated with one or more lengths of collocation involving the specified entities during the multiple collocation events.

16. The apparatus of claim 15, wherein: the at least one processing device is configured to determine the multiple determined features of the multiple collocation events for each of multiple timeframes; and the at least one processing device is configured to determine the strengths of the

different relationships using the multiple determined features of the multiple collocation events during the multiple timeframes.

17. The apparatus of claim 14, wherein: the at least one processing device is further configured to scale feature values of at least one of the multiple determined features to generate scaled features of the at least one of the multiple determined features; and the at least one processing device is configured to determine the strengths of the different relationships using the scaled feature values of the at least one of the multiple determined features.

18. The apparatus of claim 17, wherein the scaled feature values of all of the determined features span a common range of values.

19. The apparatus of claim 17, wherein: a first determined feature of the multiple determined features uses values below a first threshold to identify weaker associations involving the multiple entities and values above the first threshold to identify stronger associations involving the multiple entities, wherein a strength of an association reflects a quantification of a grouping of entities of the multiple entities; a second determined feature of the multiple determined features uses values below a second threshold to identify the stronger associations involving the multiple entities and values above the second threshold to identify the weaker associations involving the multiple entities; and the at least one processing device is configured to scale the feature values of the first determined feature or the second determined feature to reverse values representing the stronger associations involving the multiple entities and the weaker associations involving the multiple entities.

20. The apparatus of claim 14, wherein, to identify the strengths of the different relationships, the at least one processing device is configured, for each of the different relationships, to determine an aggregation of feature values associated with entities among multiple entities involved in a relationship.

21. The apparatus of claim 14, wherein the second machine learning model is trained during semi-supervised or supervised learning using labeled training data.

22. The apparatus of claim 14, wherein the second machine learning model is trained during unsupervised learning to perform anomaly detection.

23. The apparatus of claim 14, wherein: to identify the strengths of the different relationships, the at least one processing device is configured, for each of the different relationships, to determine an aggregation of feature values associated with multiple entities involved in a relationship; and unequal weights are associated with feature values of each determined feature of the multiple collocation events.

24. The apparatus of claim 14, wherein at least one of the multiple determined features is associated with distances traveled by one or more of the multiple entities to or from one or more of the multiple collocation events.

25. The apparatus of claim 14, wherein the at least one processing device is configured to use one or more statistical calculations to identify the strengths of the different relationships.

26. A non-transitory computer readable medium storing computer readable program code that when executed causes one or more processors to: obtain information associated with multiple collocation events from multiple data sources, wherein each collocation event is associated with an occurrence where multiple entities were collocated with one another; pre-process the information from the multiple data sources into a unified data format using a first machine learning model, the unified data format defining, for each collocation event: the multiple entities associated with the collocation event; a duration of the collocation event; one or more locations associated with the collocation event; and a time associated with the collocation event; determine, by a second machine learning model, multiple features of the multiple collocation events using the information in the unified data format, wherein the second machine learning model is distinct from the first machine learning model; provide the multiple determined features of the multiple collocation events to a third machine learning model, wherein the third machine learning model is distinct from the first

machine learning model and the second machine learning model; and identify strengths of different relationships involving different ones of the multiple entities using the third machine learning model based on the multiple determined features of the multiple collocation events; wherein the third machine learning model is trained to identify, using the information in the unified data format, the strengths of the different relationships by generating scores based on the multiple determined features of the multiple collocation events.

27. The non-transitory computer readable medium of claim 26, wherein the multiple determined features of the multiple collocation events comprise: one or more count features associated with one or more frequencies or numbers of collocation events involving specified entities of the multiple entities; one or more distance features associated with one or more distances between or among the specified entities during the multiple collocation events; and one or more duration features associated with one or more lengths of collocation involving the specified entities during the multiple collocation events.

28. The non-transitory computer readable medium of claim 27, wherein: the computer readable program code when executed causes the one or more processors to determine the multiple determined features of the multiple collocation events for each of multiple timeframes; and the strengths of the different relationships are based on the multiple determined features of the multiple collocation events during the multiple timeframes.

29. The non-transitory computer readable medium of claim 26, wherein: the non-transitory computer readable medium further stores computer readable program code that when executed causes the one or more processors to scale feature values of at least one of the multiple determined features to generate scaled features of the at least one of the multiple determined features; and the strengths of the different relationships are based on the scaled feature values of the at least one of the multiple determined features.

30. The non-transitory computer readable medium of claim 29, wherein the scaled feature values of all of the determined features span a common range of values.

31. The non-transitory computer readable medium of claim 29, wherein: a first determined feature of the multiple determined features uses values below a first threshold to identify weaker associations involving the multiple entities and values above the first threshold to identify stronger associations involving the multiple entities, wherein a strength of an association reflects a quantification of a grouping of entities of the multiple entities; a second determined feature of the multiple determined features uses values below a second threshold to identify the stronger associations involving the multiple entities and values above the second threshold to identify the weaker associations involving the multiple entities; and the computer readable program code that when executed causes the one or more processors to scale the feature values comprises: computer readable program code that when executed causes the one or more processors to scale the feature values of the first determined feature or the second determined feature to reverse values representing the stronger associations involving the multiple entities and the weaker associations involving the multiple entities.

32. The non-transitory computer readable medium of claim 26, wherein the computer readable program code that when executed causes the one or more processors to identify the strengths of the different relationships comprises: computer readable program code that when executed causes the one or more processors, for each of the different relationships, to determine an aggregation of feature values associated with entities among multiple entities involved in a relationship.

33. The non-transitory computer readable medium of claim 26, wherein: the computer readable program code that when executed causes the one or more processors to identify the strengths of the different relationships comprises: computer readable program code that when executed causes the one or more processors, for each of the different relationships, determine an aggregation of feature values associated with multiple entities involved in a relationship; and unequal weights are associated with feature values of each determined feature of the multiple collocation events.

34. The non-transitory computer readable medium of claim 26, wherein at least one of the multiple determined features is associated with distances traveled by one or more of the multiple entities to or from one or more of the multiple collocation events.
35. The non-transitory computer readable medium of claim 26, wherein the second machine learning model is trained during semi-supervised or supervised learning using labeled training data.
36. The non-transitory computer readable medium of claim 26, wherein the second machine learning model is trained during unsupervised learning to perform anomaly detection.
37. The non-transitory computer readable medium of claim 26, wherein the computer readable program code that when executed causes the one or more processors to identify the strengths of the different relationships comprises: computer readable program code that when executed causes the one or more processors to use one or more statistical calculations to identify the strengths of the different relationships.
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